as a sparse VAR. Since the idiosyncratic component is not observed, its parameters are estimated by an ℓ_1-regularized Yule–Walker estimator built from the idiosyncratic autocovariances implied by the estimated spectral density. The same three networks as in NETS are then estimated, and they reflect only the dependence that cannot be attributed to the common factors. The method also allows the common and idiosyncratic components to be forecast separately. The authors prove the consistency of the estimators as both the number of series and the sample size grow, also under heavier-than-Gaussian tails, and implement the procedure in the R package `fnets` (Owens et al., 2024). For our purpose, FNETS is the key method, because it separates the part of the connectedness among emerging markets that is due to global factors from the part that reflects links between the markets themselves.

1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. The Argentine episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS;